

Question	Marking Guidance	Mark	Comments
4.1	1. Binary fission; 2. Replication of (circular) DNA; 3. Division of cytoplasm to produce 2 daughter cells; 4. Each with single copy of (circular) DNA;	2 max	1. Ignore reference to 'chromosome' 2. Ignore 'copy'. 4. Ignore references to number of plasmids
4.2	1. Both denatured (by high temperature); 2. Denaturation faster at 60 °C due to more (kinetic) <u>energy</u> ; 3. Breaks hydrogen/ionic bonds (between amino acids/R groups); 4. Change in shape of the active site/active site no longer complementary so fewer enzyme-substrate complexes formed/substrate does not fit;	4	3. Ignore references to disulphide bonds 4. Accept '(at 60° C) Change in shape of the active site/active site no longer complementary so no enzyme-substrate complexes formed/substrate does not fit;
4.3	1. To digest protein; 2. (So) they can absorb <u>amino acids</u> for growth/reproduction/protein synthesis/synthesis of named cell component; OR (So) they can destroy a toxic substance/protein;	2	1. For 'digest' accept 'break down' here. 2. Accept '(so) they can destroy antibodies/antibiotics/viral antigens/bacterial antigens'

4.4	1. <u>Hydrolyse</u> (peptide bonds) to release amino acids; 2. Amino acids can cross (cell) membrane; OR Dipeptides cannot cross (cell) membrane; OR Maintain concentration gradient of amino acids for absorption; OR Ensure (nearly) maximum yield from protein breakdown;	2	2. Ignore references to crossing gut membranes. 2. Accept 'there are carrier proteins for amino acids' 2. Accept 'no carrier proteins for dipeptides'
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